Name - Abhishek Sharma
UID - 23BCS12330
Grp - KRG-3A

Input - 23
aaaaa
acacaca
aabaa
ccacacc
caaac

O/P
Yes
No
No

(514-C)

Q) Watta the owner of a spare parts store has recently got an order that can process strings in a certain way. Memory is filled with n strings initially. Then mechanism should be able to process queries of following type: "Given string S, determine if memory of mechanism contains string t that contains same no. of characters as S & differs from s in exactly at one position.

Ans) Approach -: i) Store all memory strings in set.

ii) For each query string -:
- Try changing each position
- Replace it with other two possible characters
- Check if modified string exists in set.

iii) Because -:
- only 3 characters → at each index only 2 replacements possible
- If length = L → total checks = 2L
- Total operations = O(Length)

=> Code -:

```
# include <bits/stdC++.h>
using namespace std;
int main(){
    int n,m;
    cin >>n>>m;
    unordered_set<string> st;
```

```
for(int i=0;i<n;i++){
        string s;
        cin>>s;
        st.insert(s);
}
while(m--){
        string s;
        cin>>s;
        bool found=false;
        for(int i=0;i<s.length()&&!found;i++){
                char org=s[i];
                for(char ch='a';ch<='c';ch++){
                        if(ch==org)continue;
                        s[i]=ch;
                        if(st.count(s)){
                                found=true;
                                break;
                        }
                }
                s[i]=org;
        }
        if(found)cout<<"YES\n";
        else cout<<"NO\n";
}
return 0;
}
```

O/P :

YES

NO

NO